Legend / Arrow Types: Deterministic | Stochastic | Exogenous Forcing | State Transition | Feedback Loop

LEVEL 1: External Inputs & Model Parameters

Global Parameters: $g_0, r_{\text{resp}}, \mu_{\text{turn}}, m_{\min}, C_{\text{seed}}, K_{\text{bio}}, I_{\text{rate}}, \sigma_{\text{drift}}, \sigma_{\text{thresh}}, T_{\text{cat}}, d_{\text{cat}}, p_{\text{dist}}$

Species Traits: Niche Centers $\boldsymbol{c}_s$, Covariances $\boldsymbol{\Sigma}_s$, Seed Mass $m_{\text{seed},s}$, Range $R_s = R_{\text{scale}} m_{\text{seed},s}^{\alpha}$

External Forcing: Sinusoidal Temp $T(t) = T_{\text{mean}} + A_T \sin\left(\frac{2\pi(t+\phi)}{P}\right)$, Inflow Ratio Drift $\boldsymbol{\eta}(t) \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{drift}}^2 \mathbf{I})$

LEVEL 2: Abiotic Environment & Spatially Explicit Soil Grid ($H \times W$)

Inorganic Soil Pools $\mathbf{I}(t)$

$I_{y,x,k} \geq 0, \quad k \in \{1, \ldots, N_{\text{soil}}\}$

Available Fraction: $I^{\text{avail}}_{y,x,k} = I_{y,x,k} \cdot a_{\text{avail},k}$

Organic Soil Pools $\mathbf{O}(t)$

$O_{y,x,k} \geq 0, \quad k \in \{1, \ldots, N_{\text{soil}}\}$

Recycled Inputs: $F^{\text{fresh}}_{y,x,k}$

Spatial Forcing Fields

Persistent Noise $E_{y,x} \in [0,1]$, Shock Field $S_{y,x} \in [0,1]$

Local Temp $T_{y,x} = \text{clip}\left(T(t) + \gamma(E_{y,x} - 0.5), 0, 1\right)$

Soil Nutrients I^{avail}

LEVEL 3: Individual Agent Buffer $\mathbf{A}(t)$ ($M =$ MAX_AGENTS)

$$\mathbf{a}_i(t) = (y_i, x_i, s_i, m_i, \boldsymbol{q}_i, \alpha_i, \tau_i)^T \quad \text{for } i \in \{1, \ldots, N_{\text{agents}}\}$$

$(y_i, x_i) \in [0, H) \times [0, W)$: Spatial Position

$m_i \geq 0$: Individual Plant Biomass

$s_i \in \{0, \ldots, N_{\text{spp}} - 1\}$: Species Identity

$\boldsymbol{q}_i \in [0,1]^{N_{\text{org}}}$: Elementome Ratios ($\sum_e q_{i,e} = 1$)

$\alpha_i \in \{0, 1\}$: Alive Status Flag

$\tau_i \in \mathbb{R}_{\geq 0}$: Individual Age (steps)

Uptake Depletion ($-$)

Mineralization Feedback ($+$)

Processes 1–4: Physical & Physiological Setup

[P1] Diffusion & Inflow:
$I^{\text{diff}} = \text{Conv2D}(\mathbf{I}, K), \quad I^{\text{new}} = I^{\text{diff}} + r^{\text{in}} I_{\text{rate}}$

[P2] Env Field Autoregression:
$E_{y,x}(t) = \text{Norm}\left(\rho E_{y,x}(t-1) + (1-\rho) Z_{\text{smooth}}\right)$

[P3] Patch Disturbance:
$I^{\text{dist}}_{y,x,k} = I^{\text{new}}_{y,x,k}(1 - S_{\text{dist}} E_{y,x})$ for $d(y,x;c_y,c_x) < R_{\text{dist}}$

[P4] Niche Fitness (Mahalanobis):
$w_i = \text{clip}\left(1 - \frac{(\boldsymbol{q}_i - \boldsymbol{c}_{s_i})^T \boldsymbol{\Sigma}_{s_i}^{-1} (\boldsymbol{q}_i - \boldsymbol{c}_{s_i})}{\sigma^2_{\text{thresh}}}, 0, 1\right)$

Processes 5–7: Uptake, Limitation & Growth

[P5] Michaelis-Menten Nutrient Uptake:
$U_{i,k} = U^{\text{des}}_{i,k} \cdot \frac{I^{\text{avail}}_{y_i,x_i,k}}{I^{\text{avail}}_{y_i,x_i,k} + K_m}, \quad K_m = 0.1$

[P6] Liebig Quota & Space Constraints:
$g_i^{\text{lim}} = \min_e\left(\max\left(0, 1 - \frac{Q_{\min}}{Q_{i,e}}\right)\right), \quad f_i^{\text{space}} = \max\left(0, 1 - \frac{B_{y_i,x_i}}{K_{\text{bio}}}\right)$

[P7] Realized Growth & Respiration:
$G_i^{\text{real}} = \left(U_{i,C} + \sum_k U_{i,k}\right) g_i^{\text{lim}} f_i^{\text{space}}$
$m_i'' = m_i + G_i^{\text{real}} - m_i r_{\text{resp}}(1 + \beta_r(T_{y_i,x_i} - 0.5)) + \xi_i, \quad \xi_i \sim \mathcal{N}(0, \sigma^2_{\text{demo}})$

Toroidal Geometry ($H \times W$)

Cell (y, x) Torus Wrap

$\mathcal{N}_{y,x} = \{(y + dy, x + dx) \pmod{(H, W)}\}$

Periodic Boundary Conditions

Density Crowding (K_{bio})

Process 8: Multi-Tier Mortality Filters

Viability: $\alpha_i^{(0)} = \mathbb{I}(m_i'' > 0.01)$

Catastrophe: $P_i^{\text{cat}} = d_{\text{cat}} S_{y_i,x_i}$ if $t \pmod{T_{\text{cat}}} == 0$

Weak Disturbance: $P_i^{\text{weak}} = d_{\text{weak}} E_{y_i,x_i}(1 - w_i)^2$

Strong Disturbance: $P_i^{\text{strong}} = d_{\text{strong}} S_{y_i,x_i}(\theta_i)^2, \quad \theta_i \in [0,1]$

Processes 9–10: Recycling & Reproduction

[P9] Turnover, Recycling & Mineralization:
$L_{i,e} = \left(m_i^{\text{dead}} + m_i^{\text{alive}} \mu_{\text{turn}}\right) Q_{i,e} \longrightarrow O^{\text{tot}}_{y,x,k}$
$\Phi^{\text{min}}_{y,x,k} = O^{\text{tot}}_{y,x,k} m_{\min}\left(1 + \beta_m(T_{y,x} - 0.5)\right) \longrightarrow I_{y,x,k}(t+1)$

[P10] Dispersal & Toroidal Establishment:
$y_{\text{off}} = (y_i + \Delta y) \pmod{H}, \quad x_{\text{off}} = (x_i + \Delta x) \pmod{W}$
$P^{\text{est}} = \max\left(0, 1 - \frac{B_{y_{\text{off}},x_{\text{off}}}}{K_{\text{bio}}}\right)$ with $P^{\text{seed}} = 0.1 w_i$

LEVEL 5: Emergent System State & Logged Outputs

Biomass Grid: $B_{y,x,s}(t) = \sum_{i:s_i=s} \alpha_i m_i$ **Element Pools:** $P_e(t) = \sum_i m_i q_{i,e}$ **Dissimilarity:** $\text{EDm} = \frac{1}{E} \sum_{i,j} D_M(\boldsymbol{q}_i, \boldsymbol{q}_j) \min(p_i, p_j)$